Data Handling: Import, Cleaning and Visualisation

Exercise 4: Data import and data preparation/manipulation

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Exercises: Advanced Data Import

Exercise A

Solve the problem posed in Section 11.4.2 ("Problems") in https://r4ds.had.co.nz/data-import.html, by going step-by-step through the instructions. Hint: You will need to have the `readr` package installed and loaded (should be also installed/loaded with `tidyverse`).

Exercise B

Download the file `tv_ownership.dat` posted on StudyNet along these instructions. Save this file in your current working directory/the directory dedicated to work on these exercises. The file is an excerpt of a dataset provided by the US Federal Communications Commission (FCC) with detailed data on TV and Radio stations ownership. The exerpt is in the orignal format as provided by the FCC. Figure out how to properly read this dataset into R with the functions provided in the `tidyverse/readr`-packages. Write your solution down in an R-Script and document in this script why certain problems occur and how you solved them (using comments).

Mock Exam Questions

Part 1: TRUE/FALSE Questions

For each of the following statements, decide whether the statement is TRUE (T) or FALSE (F).

1. `for`-loops are typically used in a situation where a sequence of commands needs to be executed for a well-known number of times.

Part 2: Multiple Choice Type A

For each of the following questions/statements, decide which of the possible responses are correct. NOTE: Either ONE OR SEVERAL of the response statements can be correct. In order to get a point, all correct response statements need to be marked!

1. After executing the follwing lines of R code, which value is stored in `a[3]`?

`{r eval=FALSE} a <- c(1:3) a <- a*3`

A) `1:3`
B) `9`
C) `27`
D) `3`

Part 3: Multiple Choice Type B (only one correct!)

For each of the following questions/statements, decide which of the possible responses is correct. NOTE: For each question only ONE of the possible responses is correct. Select ONLY ONE of the response statements per question!

1. The following R code was executed on the command line and the corresponding error occurred.

`read_csv("data/file1.csv")`

`Error: 'data/file1.csv' does not exist in current working directory ('/home/user/my_projects').`

Which of the following statements is correct in this context (assuming `file.csv` actually is stored somewhere on this computer's harddrive)?

A) Given the current working directory of this R session, the relative path "data/file1.csv" is wrong and `read_csv()` therefore does not find the file.
B) `read_csv()` fails to parse `file.csv` because the wrong column delimiter is used (commas).
C) `read_csv()` is not the right R function to import `file.csv`.
D) `file.csv` is generally not a valid file name.